Integration 2

Review

1.
$$\begin{aligned} y &= 5x - 3 \\ y &= 2x + 9 \\ \therefore 5x - 3 &= 2x + 9 \\ 3x &= 12x \quad = 4 \end{aligned}$$

   Since $y = 2x + 9$, $y = 8 + 9 = 17$. So $P = (4, 17)$.

2. Let $z = x^2$. Then $x^4 + 6x^2 + 4 = z^2 + 6z + 4 = 0$. Let $f(x) = z^2 + 6z + 4$.

   When $f(x) = 0$ has 1 real root, $b^2 - 4ac = 0$. When $f(x) = 0$ has 2 real roots, $b^2 - 4ac > 0$.

$$a = 1b = 6c = 4$$

   Then

$$\begin{aligned} b^2 - 4ac &= 6^2 - 4 \times 1 \times 4 \\ &= 36 - 16 \\ &= 20 \end{aligned}$$

   So $x^4 + 6x^2 + 4 = 0$ has 2 real roots.

3.
$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{x}} \\ &= x^{-\frac{1}{2}} \therefore y = 2x^{\frac{1}{2}} + c \\ &= 2\sqrt{x} + c \end{aligned}$$

   The curve passes through the point $(4, 3)$. Then

$$\begin{aligned} y &= 2\sqrt{x} + c \\ 3 &= 2\sqrt{4} + c = 4 + c \end{aligned}$$

   So $c = -1$. Therefore, $y = 2\sqrt{x} - 1$.

4. i. (d) $y = f(2x)$
   ii. (c) $y = 2f(x)$

5. i. $y = x^3 - x + 3$. Then

$$\frac{dy}{dx} = 3x^2 - 1$$

   which gives the gradient for any point on the x-axis. Then at $x = 1$,

$$\nabla t = 3(1)^2 - 1 = 2.$$

   From the general equation for a line, we get that

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 3 &= 2(x - 1) \\ y &= 2x + 1 \end{aligned}$$

   which is the equation of the tangent to the curve at the point $(1, 3)$.

ii. From i. we see that the gradient at $(1, 3)$ is 2. Then the normal to the curve will have the negative reciprocal of $2 = -\frac{1}{2}$.

$$y - y_1 = m(x - x_1)$$
$$y - 3 = -\frac{1}{2}(x - 1)$$
$$y = -\frac{1}{2}x + \frac{7}{2}$$

which is the equation of the normal to the curve at the point $(1, 3)$.

Consolidation

1. i. $\int_{-1}^{1} (4x + 5)dx$

   $= \left[2x^2 + 5x\right]_{-1}^{1}$

   $= (2 + 5) - (2 - 5)$

   $= 10$

   ii. $\int_{-1}^{0} (6x^2 - 2x)dx$

   $= \left[2x^3 - x^2\right]_{-1}^{0}$

   $= (0) - (-2 - 1)$

   $= 3$

2. i. $y = x(x - 1)$ has roots at $x = 0$ and $x = 1$. So the area between $x = 0$ and $x = 1$ is:

$$\int_{0}^{1} (x^2 - x)dx = \left[\frac{1}{3}x^3 - \frac{1}{2}x^2\right]_{0}^{1} = \left(\frac{1}{3} - \frac{1}{2}\right) - (0) = -\frac{1}{6}$$

   And the area between $x = 1$ and $x = 2$ is:

$$\int_{1}^{2} (x^2 - x)dx$$
$$= \left[\frac{1}{3}x^3 - \frac{1}{2}x^2\right]_{1}^{2}$$
$$= \left(\frac{8}{3} - 2\right) - \left(-\frac{1}{6}\right)$$
$$= \frac{5}{6}$$

   So the total area = $\left|-\frac{1}{6}\right| + \left|\frac{5}{6}\right| = 1$

3. i. sketch this TODO TODO TODO

ii.
$$\int_{-2}^{0}(x^3-x^2-6x)dx$$
$$=\left[\frac{1}{4}x^4-\frac{1}{3}x^3-3x^2\right]_{-2}^{0}$$
$$=\frac{16}{3}$$

$$\int_{0}^{3}(x^3-x^2-6x)dx$$
$$=\left[\frac{1}{4}x^4-\frac{1}{3}x^3-3x^2\right]_{3}^{0}$$
$$=\frac{63}{4}$$

So the total area is $\frac{63}{4}+\frac{16}{3}=\frac{253}{12}$

4. a.
$$\int_{1}^{2}\left(\frac{6}{x^2}-\frac{k}{x^3}\right)dx$$
$$=\left[-\frac{6}{x}+\frac{k}{2}x^2\right]_{1}^{2}$$
$$=\left(-\frac{6}{2}+\frac{k}{8}\right)-\left(-6+\frac{k}{2}\right)$$
$$=-3+\frac{k}{8}+6-\frac{k}{2}$$
$$=3-3\frac{k}{8}$$

b.
$$3-3\frac{k}{8}=0$$
$$\frac{k}{8}=1k=8$$

5. $x^3-ax^2=x^2(x-a)$ so roots at 0 and a.
$$\int_{0}^{a}(x^3-ax^2)dx$$
$$=\left[\frac{1}{4}x^4-\frac{1}{3}ax^3\right]_{0}^{a}$$
$$=\left(\frac{1}{4}a^4-\frac{1}{3}a^4\right)-(0)=-\frac{1}{12}a^4$$

Exam Questions

1.
$$\begin{aligned}y&=x^3-x^2-2x\\&=x(x^2-x-2)\\&=x(x-2)(x+1)\end{aligned}$$

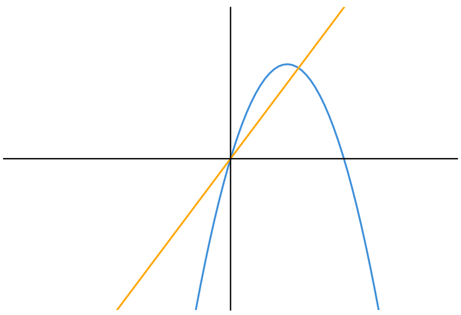
So roots at $x=-1$, $x=0$, and $x=2$. Then

$$
\begin{aligned}
&\int_{-1}^{0} (x^3 - x^2 - 2x) dx \\
&= \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - x^2\right]_{-1}^{0} dx \\
&= \frac{5}{12}
\end{aligned}
$$

And

$$
\begin{aligned}
&\int_{2}^{0} (x^3 - x^2 - 2x) dx \\
&= \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - x^2\right]_{2}^{0} dx \\
&= \frac{8}{3}
\end{aligned}
$$

So total area enclosed by the curve and the x-axis is $\frac{5}{12} + \frac{8}{3} = \frac{37}{12}$.

2. 

$$
\begin{aligned}
5x - x^2 &= 2x \\
x^2 - 3x &= 0
\end{aligned}
$$

So intersection at $x = 0$ and $x = 3$

For $y = 5x - x^2$:

$$
\begin{aligned}
&\int_{0}^{3} (5x - x^2) dx \\
&= \left[\frac{5}{2}x^2 - \frac{1}{3}x^3\right]_{0}^{3} \\
&= \frac{27}{2}
\end{aligned}
$$

For $y = 2x$:

$$
\begin{aligned}
&\int_{0}^{3} (2x) dx \\
&= \left[x^2\right]_{0}^{3} \\
&= 9
\end{aligned}
$$

Then the area of the region is $\int_0^3 (5x - x^2) dx - \int_0^3 (2x) dx = \frac{27}{2} - 9 = \frac{9}{2}$.